

PROBLEM STATEMENT #02 | Campus & Academic Operations

Automated Rubric Assignment Evaluator

1. Problem Context & Overview

Academic evaluators require an automated workflow to process batch code submissions, execute syntax and test suites, run rubric-based scoring breakdowns, and assign peer reviews without manual distribution overhead.

Target Stakeholders / Actors: *Student, Faculty Evaluator*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall automatically queue uploaded student project zip files, execute pre-configured unit test suites, and generate an itemized rubric score breakdown.

Sample Acceptance Criteria: Pass: Rubric breakdown is rendered with test results within 10 seconds of submission;

Fail: File queue stalls or invalid rubric tally.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: The system shall handle up to 100 concurrent project submissions without dropping queue items or exceeding 1.5 GB memory footprint.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.